



DSP Integrated Circuits

Chapter 2: Review of DSP Basic Theory Part I

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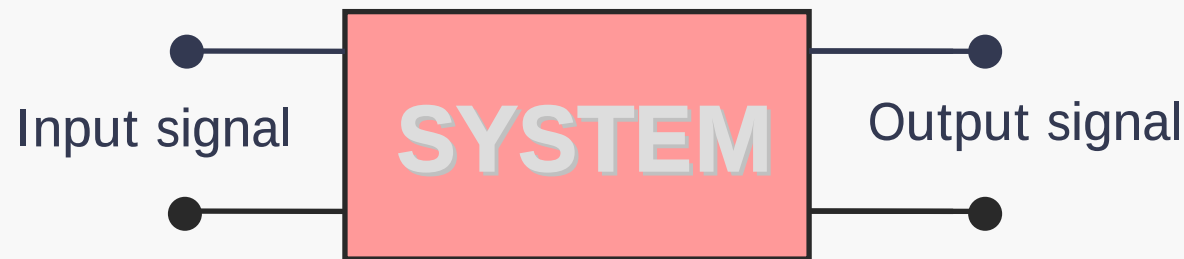


Signals and Systems

- ◇ Definition of Signals and Systems
- ◇ Linear Time Invariant (LTI) Systems

Definition of Signals & Systems

- ◇ Signal : Function of one or more independent variable
- ◇ System : Any process that results in the transformation of signals, modification of independent variable of signal



Signal Types

◇ Analog Signal :

- Varies continuously both over a range of signal values and in time

- $y = x(t)$

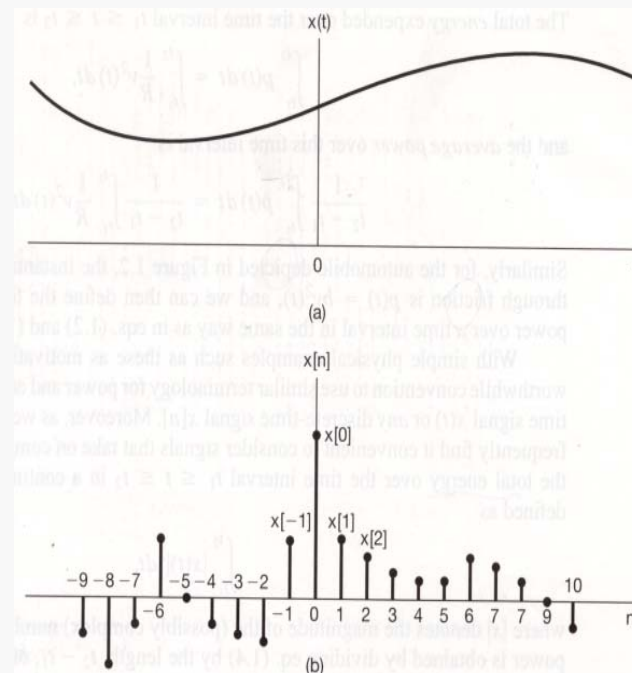
◇ Discrete-time (sampled) Signal :

- A sampled signal at discrete-time with continuously varying signal values

- $y = x(nT) = x[n]$
where T : sample period

◇ Digital Signal :

- a signal which is restricted to a discrete set of values (usually binary 0,1)



Basic Signals

	Continuous-time Signal	Discrete-time Signal
Sinusoidal Signal	$x(t) = A \cdot \cos(\omega_0 t + \phi)$	$x[n] = A \cdot \cos(\omega_0 n + \phi)$
Periodic complex exponential Signal	$x(t) = e^{j\omega_0 t}$ Where ω_0: fundamental freq.	$x[n] = e^{j\omega_0 n}$
Harmonically related set of complex exponentials	$\phi_k(t) = e^{jk\omega_0 t}, k = 0, \pm 1, \pm 2, \dots$ Where $\omega_0 = \frac{2\pi}{T_0}$	$\phi_k[n] = e^{jk\omega_0 n}, k = 0, \pm 1, \pm 2, \dots$ Where $\omega_0 = \frac{2\pi}{N}$

* Euler's relation : $e^{j\omega_0 t} = \cos \omega_0 t + j \sin \omega_0 t$



Basic operations for D-T signal

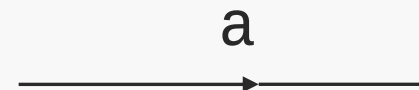
1) Summation(superposition)

$$y[n] = x_1[n] + x_2[n]$$



2) Multiplication(scaling)

$$y[n] = ax[n]$$



3) Delay(shift)

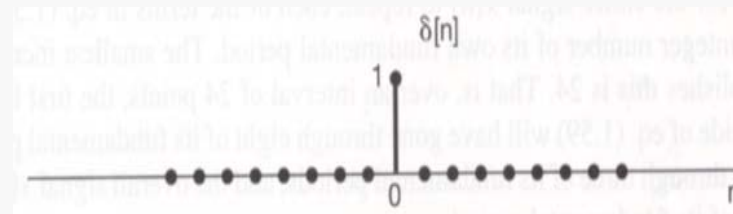
$$y[n] = x[n - d]$$



Basic Sequences

◇ Impulse Sequence :

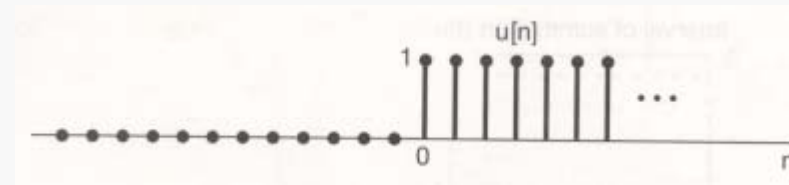
$$\delta(n) = \begin{cases} 1, & \text{if } n=0 \\ 0, & \text{otherwise} \end{cases}$$



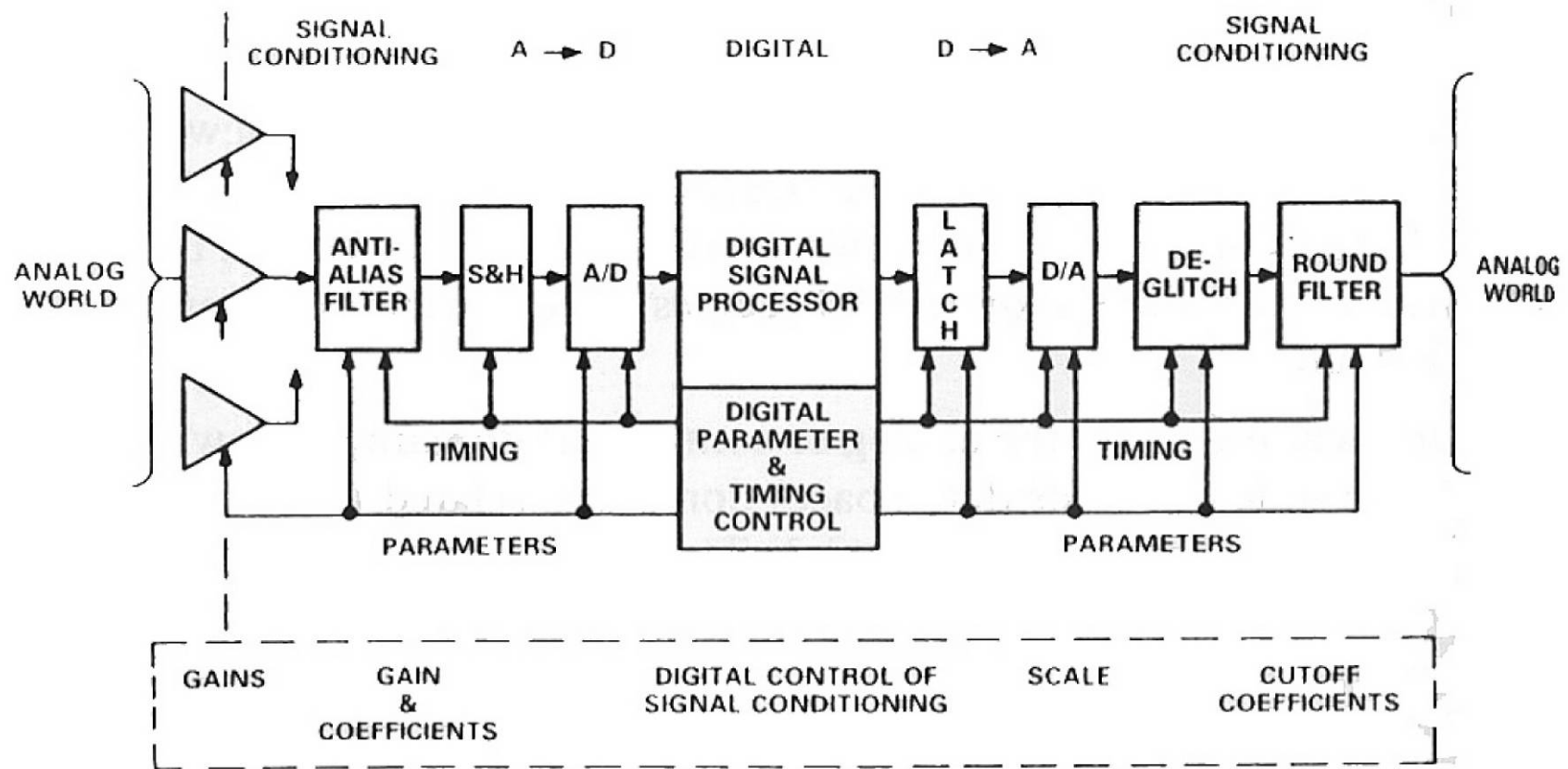
◇ Unit Step Sequence :

$$u(n) = \begin{cases} 1, & \text{if } n \geq 0 \\ 0, & \text{if } n < 0 \end{cases}$$

$$= \sum_{k=0}^{\infty} \delta[n - k]$$



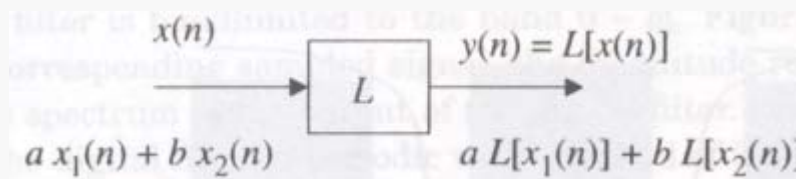
Key-elements of a sampled-data system



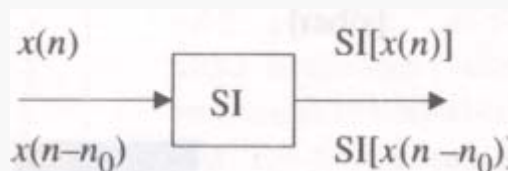
Linear Time-Invariant (LTI) System (1)

- Two basic system properties which play a fundamental role in signal and system analysis :

1) **Linear System** : If two input signals $x_1(n)$ and $x_2(n)$ yield output signals $y_1(n)$ and $y_2(n)$, the system is linear if $ax_1(n) + bx_2(n) \rightarrow ay_1(n) + by_2(n)$ for all a, b



2) **Time (Shift) -Invariant System** : If $x(n) \rightarrow y(n)$, the system is time-invariant if $x(n-n_0) \rightarrow y(n-n_0)$ for all n_0



LTI System (2)

- ◇ LTI(or LSI) system possesses superposition property :

- If $x[n] = \sum_k a_k \cdot x_k[n] \rightarrow y[n] = \sum_k a_k y_k[n]$

- ◇ LTI system shows convolution sum property :

- Output signal is determined by convolution of input signal and the impulse response of system

- $y(n) = \sum_{k=-\infty}^{\infty} x(k) \cdot h(n-k) = x(n) * h(n)$

where $h(n)$: impulse response of the system

- $Y(z) = X(z) \cdot H(z)$ in Z-domain

- ◇ Changing order of two LTI systems :

- Impulse response of a cascade of two LTI systems does not depend on the order



LTI System (3)

◇ Causal System :

- The system is causal, if output signal, $y(n_0)$ at the time instant n_0 , depends only on the input signal for time values $n \leq n_0$
- $h[n] = 0$ for $n < 0$

◇ Stable LTI System :

- LTI system is stable in bounded-input bounded output sense if $\sum_{h=-\infty}^{\infty} |h(n)| < \infty$
- The poles of a stable LTI system are confined within the unit circle in Z-plane





Discrete-Time Fourier Theory & Properties

Frequency-Domain Representation of LTI D-T Signal and Systems



input $x[n] = e^{j\omega n}$

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k] = \sum_k h[k]e^{j\omega(n-k)} = x[n]\left(\sum_k h[k]e^{-j\omega k}\right) \\ = x[n]H(\omega)$$

Frequency response (eigenvalue)

$$H(\omega + 2\pi) = \sum_k h[k]e^{-j(\omega+2\pi)k} = \sum_k h[k]e^{-j\omega k}(e^{-j2\pi})^k = H(\omega)$$

Period: 2π



Frequency-Domain Representation of LTI D-T Signal and Systems

All signals can be represented by the linear combinations of complex exponentials

$$x[n] = \sum_k a_k e^{j\omega_k n}$$

$$y[n] = \sum_k a_k e^{j\omega_k n} H(\omega_k)$$

$$\text{(ex) } x[n] = A \cos(\omega_o n + \varphi) = \frac{A}{2} (e^{j\omega_o n} e^{j\varphi} + e^{-j\omega_o n} e^{-j\varphi})$$

$$= \frac{A}{2} e^{j\varphi} e^{j\omega_o n} + \frac{A}{2} e^{-j\varphi} e^{-j\omega_o n}$$

$$y[n] = \frac{A}{2} e^{j\varphi} H(\omega_o) e^{j\omega_o n} + \frac{A}{2} e^{-j\varphi} \underbrace{H(-\omega_o)}_{= H(2\pi - \omega_o)} e^{-j\omega_o n}$$

$$\text{if } y[n] = x[n - n_d]$$

$$h[n] = \delta[n - n_d]$$

$$x[n] = e^{j\omega n} \text{ then}$$

$$\Rightarrow y[n] = x[n] e^{-j\omega n_d} H(\omega)$$

$$\text{이를 식에 넣으면 } y[n] = A \cos\{\omega_o(n - n_d) + \varphi\}$$



Discrete-Time Fourier Transform (1)

- ◇ Suitable for analysis of frequency properties of an aperiodic signal, $x[n]$
- ◇ Fourier transform of a discrete-time signal :

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad \Leftarrow \text{Freq. Response}$$

- ◇ Inverse Fourier transform :

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(\omega)e^{j\omega n} d\omega$$

- ◇ Spectral Magnitude and Phase

$$X(\omega) = X_R(\omega) + jX_I(\omega) = |X(\omega)| \angle \Phi_X(\omega) = |X(\omega)| e^{j\Phi_X(\omega)}$$

$$|X(\omega)| = \sqrt{X_R^2(\omega) + X_I^2(\omega)} \quad \Phi_X(\omega) = \tan^{-1} \left(\frac{X_I(\omega)}{X_R(\omega)} \right)$$

└→ Spectral Magnitude └→ Phase

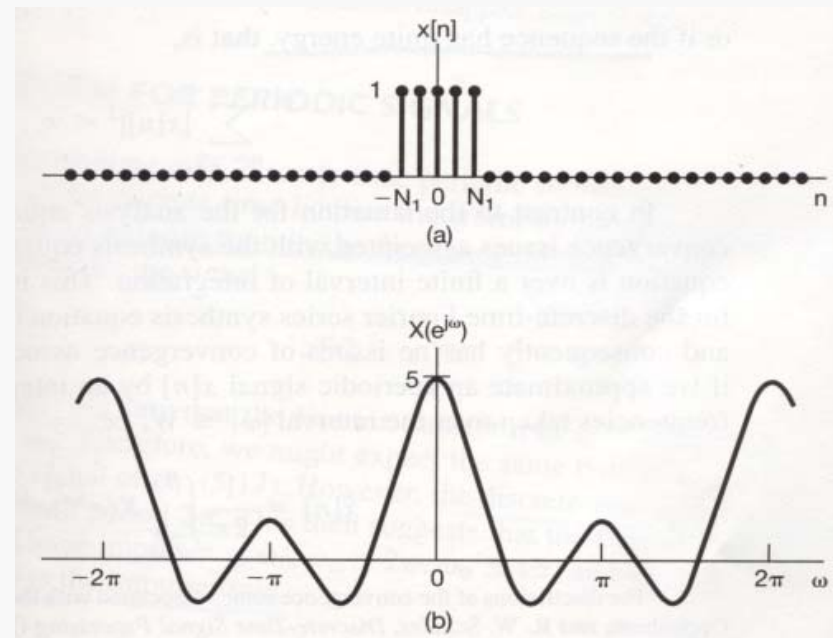


Discrete-Time Fourier Transform (2)

- ◇ Magnitude of $X(\omega)$: even function
- ◇ Phase of $X(\omega)$: odd function
- ◇ Fourier transform of rectangular pulse sequence

$$x[n] = \begin{cases} 1, & |n| \leq N_1 \\ 0, & |n| > N_1 \end{cases}$$

$$X(\omega) = \frac{\sin \omega(N_1 + \frac{1}{2})}{\sin(\omega/2)}$$



Properties of DTFT(1)

1. Linearity

$$x_3[n] = a x_1[n] + b x_2[n]$$

$$X_3(\omega) = F\{x_3[n]\}$$

$$= \sum_{n=-\infty}^{\infty} (a x_1[n] + b x_2[n]) e^{-j\omega n}$$

$$= a \sum x_1[n] e^{-j\omega n} + b \sum x_2[n] e^{-j\omega n}$$

$$= a X_1(\omega) + b X_2(\omega)$$



Properties of DTFT(2)

2. Convolution Property

$$\begin{aligned}y[n] &= h[n] * x[n] \\&= \sum_{k=-\infty}^{\infty} h[k]x[n-k] \\Y(\omega) &= \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} h[k]x[n-k]e^{-j\omega n} \\&= \sum_{k=-\infty}^{\infty} h[k] \sum_{n=-\infty}^{\infty} x[n-k]e^{-j\omega n} \\&= \sum_{k=-\infty}^{\infty} h[k] \sum_{n'=-\infty}^{\infty} x[n']e^{-j\omega n'}e^{-j\omega k} \\&= X(\omega) \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k} \\&= X(\omega) \cdot H(\omega)\end{aligned}$$



Properties of DTFT(3)

3. Modulation(windowing) property

$$\hat{h}[n] = w[n]h[n]$$

$$\hat{H}(\omega) = \sum_{n=-\infty}^{\infty} w[n]h[n]e^{-j\omega n}$$

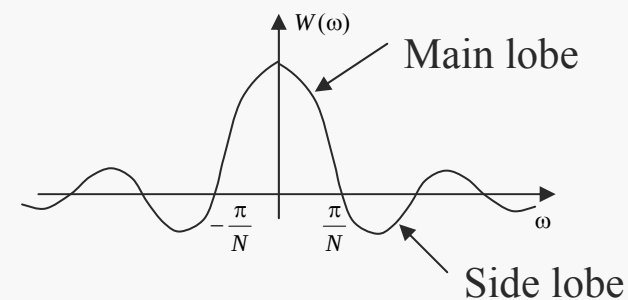
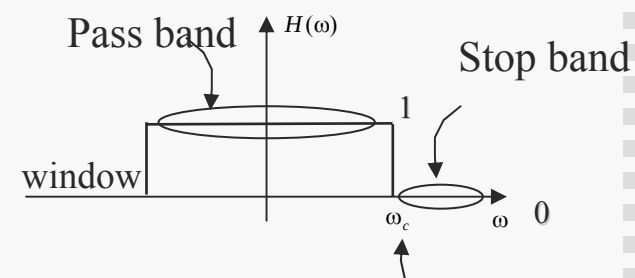
$$h[n] = \frac{1}{2\pi} \int_{2\pi} H(\omega)e^{j\omega n} d\omega$$

$$= \sum_{n=-\infty}^{\infty} w[n] \cdot \left(\frac{1}{2\pi} \int_{2\pi} H(\theta)e^{j\theta n} d\theta \right) e^{-j\omega n}$$

$$= \frac{1}{2\pi} \int_{2\pi} H(\theta) \cdot \sum_{n=-\infty}^{\infty} w[n]e^{-j(\omega-\theta)n} d\theta$$

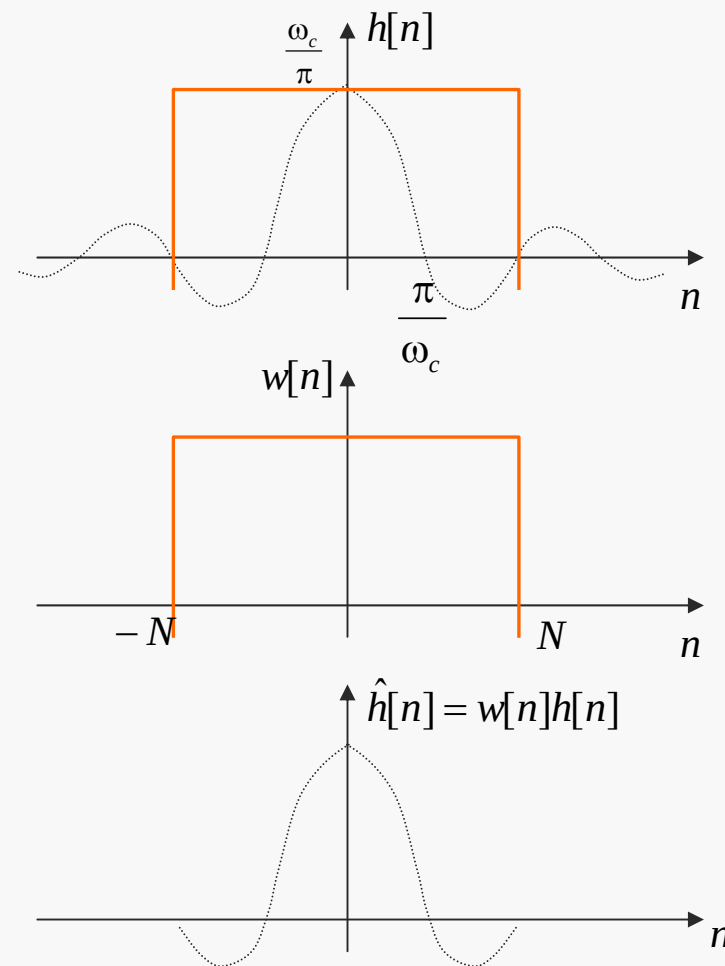
$$= \frac{1}{2\pi} \int_{2\pi} H(\theta) \cdot W(\omega - \theta) d\theta$$

$$= \frac{1}{2\pi} H(\omega) * W(\omega) = \hat{H}(\omega)$$

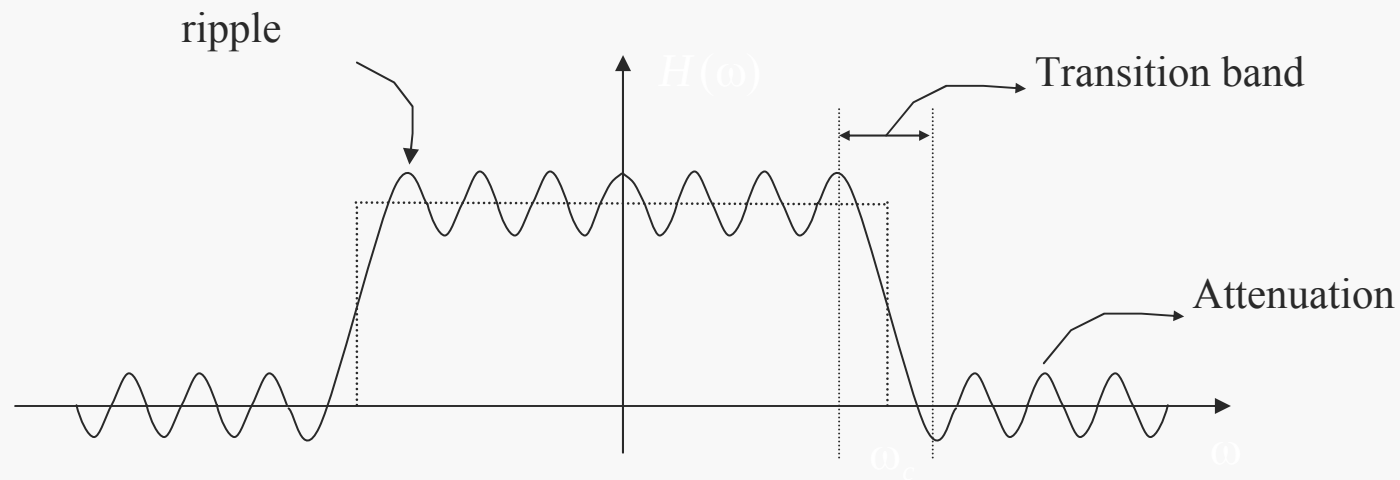


Properties of DTFT(4)

$$\begin{aligned}h[n] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} H(\omega) e^{j\omega n} d\omega \\&= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega n} d\omega \\&= \frac{1}{2\pi} \cdot \frac{1}{jn} \left[e^{j\omega_c n} - e^{-j\omega_c n} \right] \\&= \frac{\sin(\omega_c n)}{\pi n}\end{aligned}$$



Properties of DTFT(5)



Window design goal

- Transition band 를 좁힌다
- Ripple 을 되도록 작게 만든다.
- Attenuation 은 0에 가까울수록 좋다

Properties of DTFT(6)

4. Time-shifting property

$$h[n] \xleftrightarrow{FT} H(\omega)$$

$$\begin{aligned} h[n-M] &\leftrightarrow \sum_{n=-\infty}^{\infty} h[n-M] e^{-j\omega n} \\ &= \sum_{n'=-\infty}^{\infty} h[n'] e^{-j\omega(n'+M)} \\ &= \sum_{n=-\infty}^{\infty} h[n] e^{-j\omega n} \cdot e^{-j\omega M} \\ &= H(\omega) \cdot e^{-j\omega M} \end{aligned}$$

$$|H'(\omega)| = |H(\omega) \cdot e^{-j\omega M}| = |H(\omega)|$$

$$\begin{aligned} x[n] \\ X(\omega) \end{aligned}$$

$$\begin{aligned} h[n-M] \\ e^{-j\omega M} H(\omega) \end{aligned}$$

$$\begin{aligned} y[n] \\ Y(\omega) \end{aligned}$$

$$Y(\omega) = X(\omega) \cdot e^{-j\omega M} H(\omega)$$

$$= e^{-j\omega M} \underbrace{X(\omega) H(\omega)}_{Y_i(\omega)}$$



Properties of DTFT(7)

5. Differentiation in Frequency

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

$$\frac{dX(\omega)}{d\omega} = \sum_{n=-\infty}^{\infty} x[n] \cdot (-jn)e^{-j\omega n}$$

$$j \frac{dX(\omega)}{d\omega} = \sum_{n=-\infty}^{\infty} (n \cdot x[n])e^{-j\omega n}$$

$$n x[n] \stackrel{F}{\leftrightarrow} j \frac{dX(\omega)}{d\omega}$$

6. Parseval's Theorem

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{2\pi} |X(\omega)|^2 d\omega$$

